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In the paper “*Marangoni Spreading on Liquid Substrates in New Media Art*” by San To Chan and Eliot Fried (Chan & Fried, 2024) conduct a research experiment focusing on Marangoni spread in visual art shows by Japanese artist Akiko Nakayama (Beppu Galleria Midobaru, 2021). Some background on the art made by the artist Akiko Nakayama, is that Nakayama creates art by making live performance shows that use a mixture of alcohol and water to make lives shows that demonstrate her art.



Figure 1: One of Akiko Nakayama live art performances. (Beppu Galleria Midobaru, 2021)



Figure 2: Example of a finished painting by Akiko Nakayama. (Beppu Galleria Midobaru, 2021)

The art utilizes forces that are commonly explored in engineering. The experiment seeks to replicate her art demonstrations and explore the relation between dendritic painting and the forces that are a part of chemical engineering such as fluid mechanics, rheology, and statistical physics (Chan & Fried, 2024). The importance of expanding this relation between art and forces behind engineering is that both can mutually benefit each other if research is conducted behind the forces that govern the phenomena (Perkowitz, 2024). To start with the relation, we can look into how Marangoni spread is important with engineering, as the application applies very well to paints and coating of surfaces. The spread of the paints or the coaters are impacted on how Marangoni forces move around the surface, this effect has a lot of research and development behind it as there is always a need to improve spreading and distribution on various surfaces. In art specifically in dendritic painting Marangoni forces are the force that determines how art looks. By relating these two topics, that being art and engineering, is how the art uses the forces on a smaller scale and has the same governing forces that are explored in engineering. The art can be used to simulate the effect of the spread that can later be implemented into the design of materials in engineering.

Meanwhile all the research focused on materials that can make more intrinsic patterns to help make art. These applications are very important as the industry in paint and coating is always for ways to effectively spread and maintain a constant layer on a surface while reducing cost and material usage. Such an example being the thermal coating industry where the primers and can coats in thermal isolation coating materials are very expensive and difficult to spread due to the material properties of surface applied and the coating itself. With further understanding of how these forces work and how they can be used to improve coating a lot of research can be focused on improving the primers by understanding Marangoni forces. (Thermal Spray, n.d.)

Now onto the problem statement this research reveals how fluid mechanics and material rheology govern the formation of dendritic patterns in contemporary painting techniques. By studying how ink and alcohol droplets spread across liquids with different flow properties, the authors identified universal scaling laws driven by Marangoni forces and connected the resulting structures to concepts from statistical physics such as kinetic roughening and diffusion limited aggregation. The work demonstrates that artistic pattern formation can serve as a model system for studying complex interfacial instabilities in soft matter physics. Beyond art, these findings may improve technologies involving liquid spreading, coating, printing, and surface engineering. The study also highlights how nonlinear fluid behavior can generate highly organized structures from simple physical interactions.

To start the critique of the paper “*Marangoni Spreading on Liquid Substrates in New Media Art*” by San To Chan and Eliot Fried (Chan & Fried, 2024). We first must understand how the forces that are researched, the main forces in the paper that are discussed are Marangoni forces, viscous forces, gravitational effects, and nonlinear rheological forces, which together govern the spreading, instability growth, and dendritic pattern formation of ink/alcohol droplets on liquid

substrates. All of which when combined are competing with one another and lead to the dendritic painting patterns made by Akiko Nakayama live art performances.

To start with we Marangoni forces are the primary driving mechanism behind the spreading behavior observed in the experiments. These forces arise from differences in surface tension between the ink/alcohol droplet and the surrounding liquid substrate, causing fluid to flow from regions of low surface tension toward regions of higher surface tension. Theoretically, this behavior is described through a balance between Marangoni stress and viscous stress, producing characteristic spreading laws such as (de Gennes et al., 2004):

$$R \sim t^{3/8}$$

and

$$R \sim t^{1/4}$$

These scaling relations demonstrate how surface tension gradients govern the outward expansion of the droplet and the development of dendritic spreading patterns.

As the droplets spread, viscous forces within the liquid substrate resist the motion of the fluid and dissipate energy through internal friction. According to Newton's law of viscosity, viscous stress increases proportionally with the shear rate in Newtonian fluids, while shear thinning fluids exhibit decreasing viscosity as shear rate increases (de Gennes et al., 2004). The balance between Marangoni driven flow and viscous resistance determines the spreading rate and strongly influences the morphology of the dendrites. More viscous substrates slow the fluid motion and suppress lateral spreading, resulting in narrower and more refined branching structures.

Although gravitational effects were considered in the analysis, they were found to play a relatively minor role compared to surface tension and viscous forces. The relative importance of gravity is measured through the Bond number, which compares gravitational forces to capillary forces (de Gennes et al., 2004):

$$Bo = \frac{\rho g H^2}{\Delta\sigma}$$

Because the Bond number remained small in most experiments, the authors concluded that gravitational forces were negligible and that surface tension effects dominated the droplet spreading dynamics.

Beyond these bulk forces, the paper also discusses Laplacian growth effects, which are responsible for the branching and fractal like nature of the dendrites. Laplacian growth occurs when interface evolution is governed by solutions to the Laplace equation, causing protruding regions to grow faster than surrounding areas (Lee et al., 2017). Similar behavior is observed in diffusion limited aggregation and viscous fingering systems, where small perturbations become amplified over time. This mechanism explains the formation of selfavoiding tree like dendritic structures during Marangoni spreading.

Finally, the authors examine how nonlinear rheological behavior and kinetic roughening contribute to the evolution of the droplet interface. The interface roughness is analyzed using theories such as the Kardar Parisi Zhang (KPZ) model (Lee et al., 2017), which describes stochastic interface growth through local fluctuations and nonlinear propagation. The governing relation for KPZ roughening is:

$$\frac{\partial h}{\partial t} = v\Delta h + \frac{\lambda}{2} |\nabla h|^2 + \zeta$$

where the competing terms describe smoothing, nonlinear growth, and random fluctuations (de Gennes et al., 2004). With these theoretical models the paper can explain how microscopic disturbances evolve into large scale roughened dendritic boundaries on shear thinning substrates.

With the forces established and background knowledge on what leads to the creation of the paper and the forces explored in the paper the critique on the paper can commence on what the paper did well, how the experiment was conducted and how the experiment can relate to the theory behind the paper's development.

To start with the strengths of the paper, the paper successfully was able to relate the theory and application of the experiment to help expand in the potential in expanding on research focused on the forces that determine spread. This paper successfully connects fluid engineering mechanics and are with art through theory and visuals with spreading patterns and their governing equations that determine the forces behind it.

To start with one of the greatest strengths of the paper is its successful integration of art and science into a rigorous fluid mechanics study. Rather than treating dendritic painting solely as an artistic process, the authors demonstrate that it can also serve as a physically meaningful system for studying interfacial instabilities, Marangoni spreading, and nonlinear rheology (Chan & Fried, 2024). This approach allows for the research to be highly original and broadens its relevance beyond traditional fluid mechanics by connecting it to soft matter physics, statistical mechanics, and contemporary media art.

Another major strength of the paper is the quality and organization of the experimental methodology. The authors systematically varied important parameters such as IPA concentration,

substrate thickness, and substrate rheology while carefully measuring spreading radii, interface roughness, and dendritic morphology (Chan & Fried, 2024). The inclusion of Newtonian, shear thinning, and power law substrates allowed the experiments to isolate how rheological properties influence spreading dynamics. This structured experimental design strengthens the validity of the conclusions and allows clear comparisons between different flow regimes.

The paper is also very strong in its use of theoretical scaling laws to explain the observed spreading behavior. The authors experimentally validated the characteristic Marangoni spreading relations:

$$R = t^{3/8}$$

and

$$R = t^{1/4}$$

which had previously received limited experimental confirmation (Chan & Fried, 2024). By collapsing experimental data onto universal master curves, the study provides convincing evidence that Marangoni driven spreading governs the evolution of the droplets. The agreement between experiment and theory is one of the strongest scientific contributions of the work.

A further strength is the paper's connection between dendritic spreading and broader theories in statistical physics. The authors relate the observed interface roughening to KPZ and quenched KPZ universality classes, while also comparing the dendritic structures to diffusion limited aggregation and viscous fingering phenomena (Chan & Fried, 2024). These comparisons significantly elevate the paper because they demonstrate that the observed patterns are not isolated artistic effects, but rather examples of universal nonlinear growth behavior found throughout nature.

Visual presentation of the experiments is another important strength. The quality images of droplet spreading and dendritic formation clearly illustrate the evolution of the interfaces and make the physical phenomena easier to interpret. The progression from smooth spreading to fractal like dendrites is especially effective in supporting the theoretical analysis (Chan & Fried, 2024). Because the paper focuses on highly visual fluid phenomena, the strong imaging and figure organization greatly improve the clarity and impact of the study.

Finally, the paper is strong because it opens many opportunities for future research and practical applications. The findings may contribute to better understanding of coating technologies, printing systems, surface engineering, and fluid based manufacturing processes where interfacial instabilities are important (Chan & Fried, 2024). Additionally, the authors were able to identify several future research directions involving active fluids, viscoelastic materials, and other instability mechanisms, showing that the work provides a foundation for continued exploration in both physics and materials science that can be implemented into industry and product development.

Although the experimental design of the paper is well organized and visually effective, several limitations within the methodology may affect the accuracy and interpretation of the results. One weakness that is highly demonstrated in the paper is the lack of environmental control during the experiments. Since the spreading process is strongly driven by IPA evaporation and surface tension gradients, factors such as small fluctuations in temperature, humidity, or airflow could significantly alter the Marangoni forces responsible for dendritic growth. While the authors report performing the experiments at room temperature and controlled humidity, the setup was not fully isolated in an environmental chamber (Chan & Fried, 2024). As a result, variations in

evaporation rates may have introduced inconsistencies into the spreading dynamics and interface evolution leading to questions in the credibility of these findings.

Another limitation is the relatively small number of experimental repetitions used for some measurements. The paper states that experiments were repeated at least three times for measurements of spreading radii and six times for roughness analysis. While this is statistically reliable, dendritic growth and instability formation are highly nonlinear processes that can be sensitive to minor disturbances or imperfections in the substrate (Chan & Fried, 2024). A larger sample size would improve confidence in the reported scaling exponents and reduce uncertainty in the experimental averages. With more extensive statistical analysis that could also help determine whether some deviations from universal behavior arise from experimental variability or from actual physical transitions.

The imaging and interface detection methods may also introduce uncertainty into the results. The spreading in the radii and roughness exponents were extracted from high speed video recordings and image binarization techniques. However, dendritic interfaces become increasingly diffused and irregular at higher IPA concentrations, making the precise location of the interface difficult to determine within the experimental replications. Small variations in thresholding during image processing could influence measurements of roughness and fractal dimension, especially for the highly branched power law fluid cases (Chan & Fried, 2024). The paper does not deeply discuss sensitivity analysis for the image processing methods, leaving uncertainty regarding how robust the extracted scaling exponents truly are.

The rheological characterization of the substrates also presents some limitations. The authors modeled the acrylic resin and paint mixtures using the Cross model for shear thinning fluids, which assumes a simplified relationship between viscosity and shear rate. However, the

spreading process likely involves transient stresses, extensional flows, and localized deformation rates that are more complex than those captured through steady state rotational rheometry. Many acrylic based materials also exhibit viscoelastic and thixotropic behavior, yet these effects were not thoroughly investigated (Chan & Fried, 2024). Because interfacial instabilities are highly sensitive to rheological properties, neglecting elastic contributions may oversimplify the mechanisms responsible for dendritic formation.

Another experimental concern involves the assumption of uniform substrate thickness throughout the spreading process. Although the liquid layers were deposited using a film applicator, thin liquid films can still contain local thickness variations or surface irregularities (Chan & Fried, 2024). These small inconsistencies may act as perturbations that influence dendrite initiation and growth direction. Since the study focuses heavily on instability formation and interface roughening, even minor substrate nonuniformities could affect the resulting morphologies. Direct measurements of film thickness profiles before and during spreading would strengthen the reliability of the conclusions.

The experiments also lack direct flow field measurements that could confirm the proposed physical mechanisms. The authors infer the existence of Marangoni driven flow, Laplacian growth, and viscous fingering behavior primarily through morphological observations and scaling analysis (Chan & Fried, 2024). While these interpretations are reasonable, techniques such as particle image velocimetry, fluorescent tracer imaging, or interferometry could provide direct visualization of velocity fields and concentration gradients within the spreading droplets. Such measurements would offer stronger evidence for the proposed mechanisms and help distinguish between competing instability processes.

Finally, the experiments remain phenomenological rather than predictive. While the study is successful in identifying scaling laws and roughening behavior, it does not establish a clear predictive criterion for when dendrites will form or transition between different growth regimes. For example, the paper does not define a critical Marangoni number, Capillary number, or rheological threshold separating smooth spreading from fractal dendritic growth (Chan & Fried, 2024). Developing such predictive relationships would significantly improve the practical usefulness of the experiments for applications in coating flows, printing technologies, and fluid based manufacturing systems.

The theoretical framework presented in the paper is highly ambitious and successfully connects fluid mechanics, rheology, and statistical physics; however, several assumptions and simplifications limit the credibility of the analysis. One of the primary theoretical limitations is the reliance on simplified scaling arguments to describe the spreading behavior of the droplets. The authors derive the characteristic spreading laws by balancing Marangoni stress and viscous stress, leading to the power law relations:

$$R \sim t^{3/8}$$

and

$$R \sim t^{1/4}$$

Although these scaling laws agree well with experimental observations, they are derived under highly idealized assumptions. The theory assumes simplified force balances, constant material properties, and dominant Marangoni driven flow while neglecting many coupled transport processes that likely occur simultaneously during spreading. In reality, the system involves

evaporation, concentration gradients, diffusion, and continuously changing fluid composition, all of which may alter the validity of the scaling relations over time.

Another weakness of the theoretical analysis is the assumption that the surface activity parameter and surface tension gradients remain effectively constant throughout the spreading process.

Because IPA evaporates rapidly, the concentration within the droplet changes continuously as spreading occurs. This means the Marangoni stress itself evolves dynamically rather than remaining constant. The theory does not fully account for transient evaporation driven concentration transport or the coupling between evaporation and fluid motion. A more rigorous treatment would require solving coupled mass transfer and fluid dynamics equations simultaneously rather than relying primarily on scaling analysis.

The rheological modeling of the liquid substrates also introduces theoretical simplifications that may limit the interpretation of the results. The authors use the Cross model to describe the shear thinning behavior of the acrylic resin and paint mixtures. While this model captures the dependence of viscosity on shear rate, it neglects other potentially important rheological phenomena such as viscoelasticity, thixotropy, and normal stress differences. Many polymeric and acrylic fluids exhibit elastic stresses under deformation, especially in extensional or transient flows. These elastic effects can strongly influence interfacial instabilities and dendritic growth, yet they are not included in the theoretical framework. As a result, the theory may oversimplify the mechanisms responsible for the observed morphologies.

The interpretation of dendritic spreading as a Laplacian growth process is conceptually compelling but remains somewhat indirect. The authors connect the observed fractal structures to diffusion limited aggregation and viscous fingering phenomena through similarities in morphology and fractal dimension.(Chan & Fried, 2024) However, the study does not directly

solve the governing Laplace equation or experimentally measure the concentration fields responsible for Laplacian growth. The argument is therefore based primarily on analogy rather than direct theoretical demonstration. More rigorous mathematical modeling of the concentration and velocity fields would strengthen the claim that Marangoni spreading in this system truly behaves as a Laplacian growth process.

Another theoretical concern involves the application of KPZ and quenched KPZ universality classes to the roughening behavior of the interfaces. The authors interpret the experimentally measured roughness and growth exponents as evidence for qKPZ dynamics in the shear thinning substrates. The governing KPZ equation used in the analysis is:

$$\frac{\partial h}{\partial t} = \nu \Delta h + \frac{\lambda}{2} |\nabla h|^2 + \zeta$$

While the measured exponents show qualitative agreement with KPZ theory, the scaling regions appear relatively limited in both space and time. In statistical physics, universality class identification typically requires extensive scaling ranges and careful finite size analysis. Since the experiments are performed over relatively short timescales and confined geometries, the observed exponents may reflect transient or crossover behavior rather than true asymptotic universality. Consequently, the assignment of KPZ or qKPZ behavior may be somewhat premature without additional experimental or computational evidence.

The theoretical treatment also assumes that continuum mechanics remains valid throughout the spreading process, even within extremely thin precursor films. However, the precursor films are estimated to have thicknesses approaching micrometer scales, where intermolecular forces, evaporation effects, and microscopic heterogeneities may become increasingly important. Under such conditions, continuum assumptions may begin to break down, and molecular scale

interactions could influence the spreading dynamics. The theory does not incorporate disjoining pressure, intermolecular interactions, or other thin film effects commonly used in lubrication theory for microscale spreading systems.

Additionally, the paper primarily focuses on identifying scaling behavior rather than developing predictive governing equations for instability onset. While the scaling laws successfully describe how spreading evolves after dendrites form, the theory does not clearly predict when smooth spreading transitions into dendritic instability. Important dimensionless groups such as the Marangoni number and Capillary number are not extensively analyzed as instability criteria. Developing predictive thresholds for dendrite formation would significantly strengthen the theoretical framework and improve its applicability to engineering systems involving coatings, printing, or thin film transport.

Despite these limitations, the theoretical approach remains one of the strongest aspects of the paper because it successfully connects several complex physical phenomena into a unified framework. The combination of Marangoni spreading theory, Laplacian growth concepts, nonlinear rheology, and kinetic roughening provides a highly interdisciplinary interpretation of dendritic painting. Even though some assumptions simplify the underlying physics, the framework offers an important foundation for future studies that may incorporate more detailed numerical simulations, coupled transport models, and direct flow field measurements.

The behavior observed in dendritic painting shares similarities with several well known physical systems involving interfacial instability, pattern formation, and nonequilibrium transport phenomena. One of the closest related systems is Marangoni spreading, in which surface tension gradients drive liquid motion across an interface. Chan and Fried demonstrated that the spreading radii of ink/isopropanol droplets follow the characteristic $t^{3/8}$ and $t^{1/4}$ power laws associated

with Marangoni driven transport, confirming that surface tension gradients are the dominant driving mechanism behind the observed dendritic growth (Chan & Fried, 2024). Similar spreading behavior has previously been reported in surfactant driven thin films and liquid coating systems, where surface tension differences generate strong interfacial flows capable of destabilizing the liquid front.

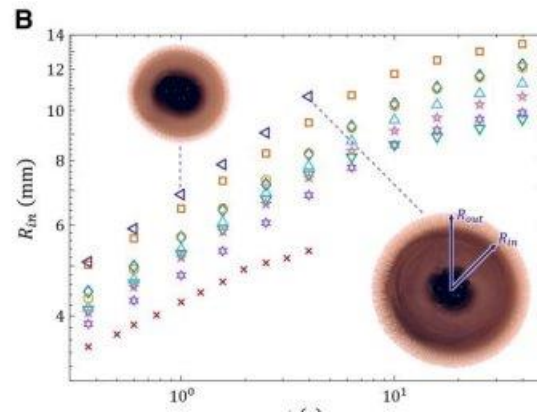


Figure : Determines the relation in the spread of inner and outer radius following the spreading factor. (Chan & Fried, 2024)

The dendritic structures produced during the experiments also resemble viscous fingering patterns commonly observed in Hele–Shaw cells and porous media flows. In viscous fingering, a less viscous fluid displaces a more viscous one, causing branching instabilities that evolve into tree like morphologies. Chan and Fried argue that the observed dendrites are analogous to this phenomenon because the low viscosity alcohol rich droplets spread across comparatively viscous acrylic substrates. The resulting instability promotes the formation of elongated branches similar to diffusion limited aggregation (DLA) structures. Their measured fractal dimension of approximately 1.68 closely matches values reported for classical DLA systems, suggesting that the spreading process belongs to a broader class of Laplacian growth phenomena (Chan & Fried, 2024).

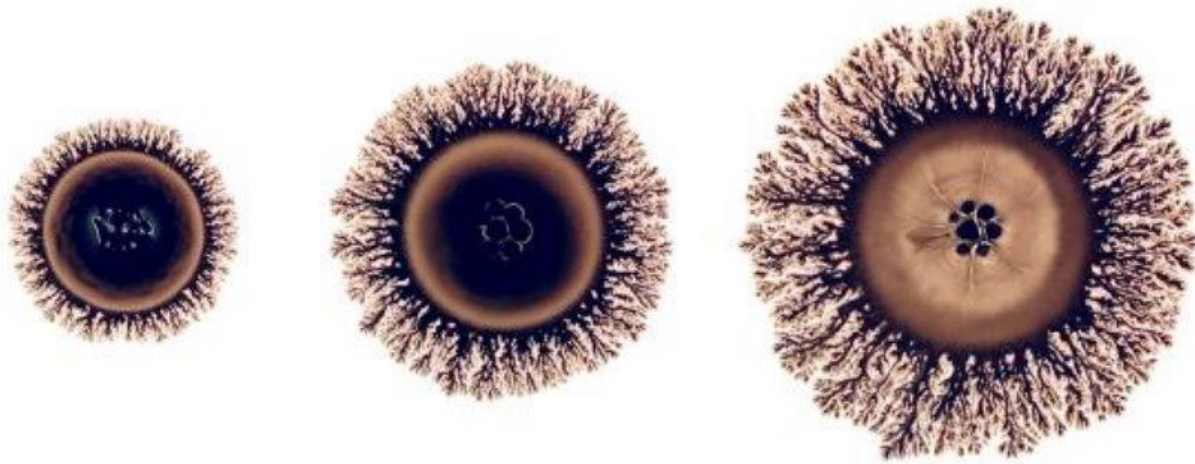


Figure : Demonstrates how the art is form when forces interact creating the patterns. (Chan & Fried, 2024)

The work also connects strongly to kinetic roughening theory and the Kardar Parisi Zhang (KPZ) universality framework. The roughened precursor film surrounding the droplets exhibited scaling exponents close to those predicted by the quenched KPZ model. Similar scaling behavior has been observed in bacterial colony growth, electrochemical deposition, and fluid invasion through porous materials. In each of these systems, local disorder and nonlinear interface propagation produce statistically self similar roughening patterns over time. The appearance of qKPZ like exponents in dendritic painting indicates that substrate rheology and microscopic structural heterogeneity strongly influence interface evolution.

Beyond fluid mechanics, these findings relate to broader studies of wetting and capillarity. In de Gennes et al, they describe how thin film spreading, wetting transitions, and surface energy minimization govern many interfacial phenomena in liquids (de Gennes et al., 2004). Likewise,

nanoscale surface interactions and intermolecular forces discussed by Lee et al. help explain how viscosity, adhesion, and molecular scale forces affect spreading dynamics and thin film stability (Lee et al., 2017). Together, these related studies demonstrate that dendritic painting represents an intersection of fluid mechanics, rheology, statistical physics, and surface science (Chan & Fried, 2024).

The paper opens several promising opportunities for future research and experimental improvement. One important direction would involve developing more advanced theoretical and computational models that fully couple evaporation, concentration transport, rheology, and fluid motion. The current study primarily relies on scaling laws and simplified force balances to explain the spreading behavior. While these models successfully capture the general trends of the experiments, they do not fully describe the transient evolution of surface tension gradients caused by IPA evaporation. Future work could incorporate coupled Navier Stokes equations with diffusion and evaporation terms to better predict the formation and evolution of dendritic structures. Such numerical simulations would help clarify the relative importance of Marangoni stresses, viscous resistance, and nonlinear rheological effects during the spreading process.

Another major improvement would involve the use of advanced flow visualization techniques to directly measure the internal dynamics of the droplets and substrate. The current experiments infer the presence of Marangoni driven flow and Laplacian growth primarily through morphological observations and scaling behavior. Techniques such as particle image velocimetry, fluorescent tracer imaging, or interferometry could provide direct measurements of velocity fields, concentration gradients, and film thickness evolution. These methods would strengthen the physical interpretation of the results and provide more direct evidence for the proposed

instability mechanisms. Additionally, higher resolution imaging systems could improve the accuracy of interface roughness and fractal dimension measurements.

Future studies could also investigate a wider range of rheologically complex materials. The paper primarily focuses on Newtonian, shear thinning, and power law fluids, but many industrial fluids exhibit viscosity, elasticity, and yield stress behavior. The studying dendritic spreading on viscous and elastic substrates may reveal entirely new instability mechanisms due to elastic stresses within the fluid. Similarly, introducing anisotropic or structured substrates could allow researchers to control the direction and morphology of dendritic growth. Such forces may have practical applications in coating technologies, printing processes, and surface engineering.

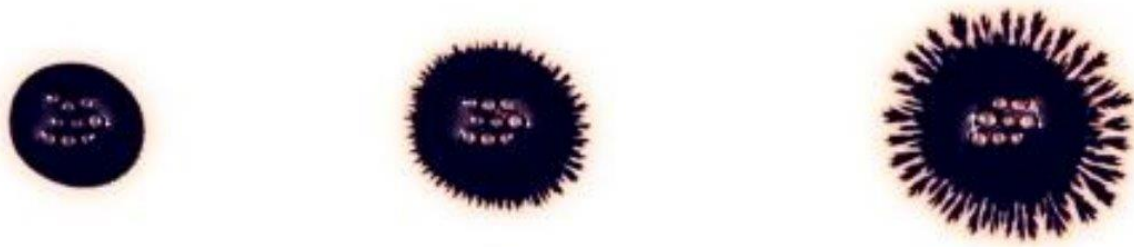


Figure : Demonstrates a small experiment on the Newtonian fluid relation on the forces. (Chan & Fried, 2024)

Another change that would be beneficial to explore is the role of environmental conditions in greater detail. Because the spreading process is strongly dependent on evaporation driven surface tension gradients, factors such as temperature, humidity, and airflow likely influence the resulting morphologies. Conducting experiments within a fully controlled environmental

chamber would improve reproducibility and allow systematic investigation of how evaporation rates affect dendritic growth. Future work could also study the effects of varying substrate thickness and wetting more precisely to determine how confinement influences the transition between smooth spreading and fractal instability formation.

The paper also creates opportunities for broader interdisciplinary applications. Since the dendritic structures resemble patterns found in diffusion limited aggregation, viscous fingering, and bacterial colony growth, future studies could compare these systems more directly using statistical and computational methods. Using machine learning techniques could potentially identify dendritic morphologies and identify hidden transitions between different growth regimes. Additionally, the principles discovered in this work may contribute to practical technologies involving fluid transport and coating flows in applications like surface coatings.

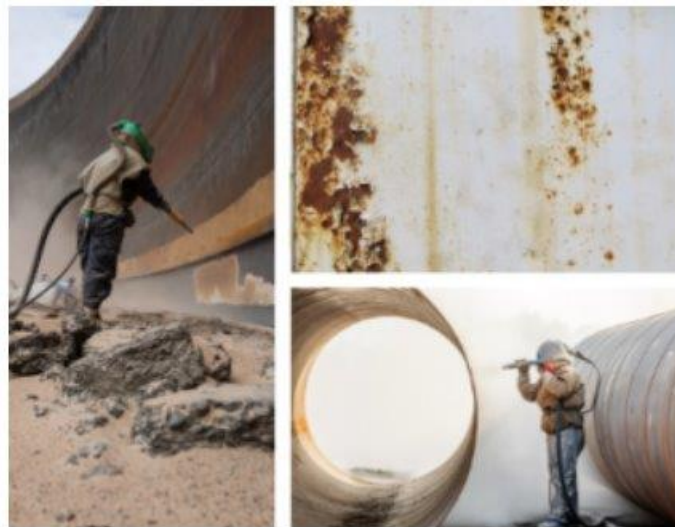


Figure : Application of coating on surface that relates to the experiment. (Thermal Spray, n.d.)

Finally, future research could focus on expanding the connection between art and physics by exploring other artistic fluid systems. The authors briefly mention watercolor salt patterns and active liquid systems as possible research directions to expand on. Investigating systems may

reveal additional focus in instability mechanisms, or nonlinear transport phenomena. Overall, the paper provides a strong foundation for future studies combining fluid mechanics, soft matter physics, nonlinear dynamics, and artistic pattern formation.

In conclusion, the paper “*Marangoni Spreading on Liquid Substrates in New Media Art*” presents an innovative and interdisciplinary investigation into the physics underlying dendritic painting. By combining experimental observations with scaling theory, rheology, and statistical physics, the authors of the paper successfully demonstrate how Marangoni forces, viscous effects, and nonlinear substrate behavior govern the formation of complex dendritic structures. The work is particularly strong in its experimental visualization, proof of Marangoni spreading laws, and its connection of artistic fluid phenomena to broader concepts such as Laplacian spread and KPZ roughening.

Despite these strengths of the experiment, several limitations remain within both the experimental and theoretical approaches. Simplifications involving evaporation, rheological modeling, and universality class identification may restrict the completeness of the conclusions. Additionally, the absence of direct flow field measurements and predictive instability criteria leaves important physical mechanisms only partially resolved. Nevertheless, the paper provides a strong foundation for future research involving complex fluids, thin film instabilities, and interfacial transport phenomena. Overall, the study successfully bridges art and science while offering meaningful contributions to fluid mechanics, soft matter physics, and nonlinear pattern formation.

References:

- Beppu Galleria Midobaru. (2021, August 21). *Akiko Nakayama*. <https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama/> (Beppu Galleria Midobaru, 2021)
- Chan, S. T., & Fried, E. (2024). *Marangoni spreading on liquid substrates in new media art*. *PNAS Nexus*, 3(2), 1–12. <https://doi.org/10.1093/pnasnexus/pgae059> (Chan & Fried, 2024)
- de Gennes, P.-G., Brochard-Wyart, F., & Quéré, D. (2004). *Capillarity and wetting phenomena: Drops, bubbles, pearls, waves*. Springer. <https://doi.org/10.1007/978-0-387-21656-0> (de Gennes et al., 2004)
- Lee, D. W., Ruths, M., & Israelachvili, J. N. (2017). Surface forces and nanorheology of molecularly thin films. In B. Bhushan (Ed.), *Nanotribology and nanomechanics* (pp. 457–518). Springer. https://doi.org/10.1007/978-3-319-51433-8_9 (Lee et al., 2017)
- Perkowitz, S. (2024, September 3). *Akiko Nakayama: The Japanese artist skilled in fluid mechanics*. *Physics World*. <https://physicsworld.com/a/akiko-nakayama-the-japanese-artist-skilled-in-fluid-mechanics/> (Perkowitz, 2024)
- Thermal Spray. (n.d.). *All you need to know about metal film coatings*. <https://www.thermalspray.com/all-you-need-to-know-about-metal-film-coatings/> (Thermal Spray, n.d.)